AQUAFAB INDUSTRIES

Commissioning a New Machine

Checking the hardware, then calibrating flow, temperature and oxygen.

For Aquafab Workbench 0.2.0

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Screenshots show simulated machines and modelled example data, not measurements. Aquafab Workbench is not a safety system.

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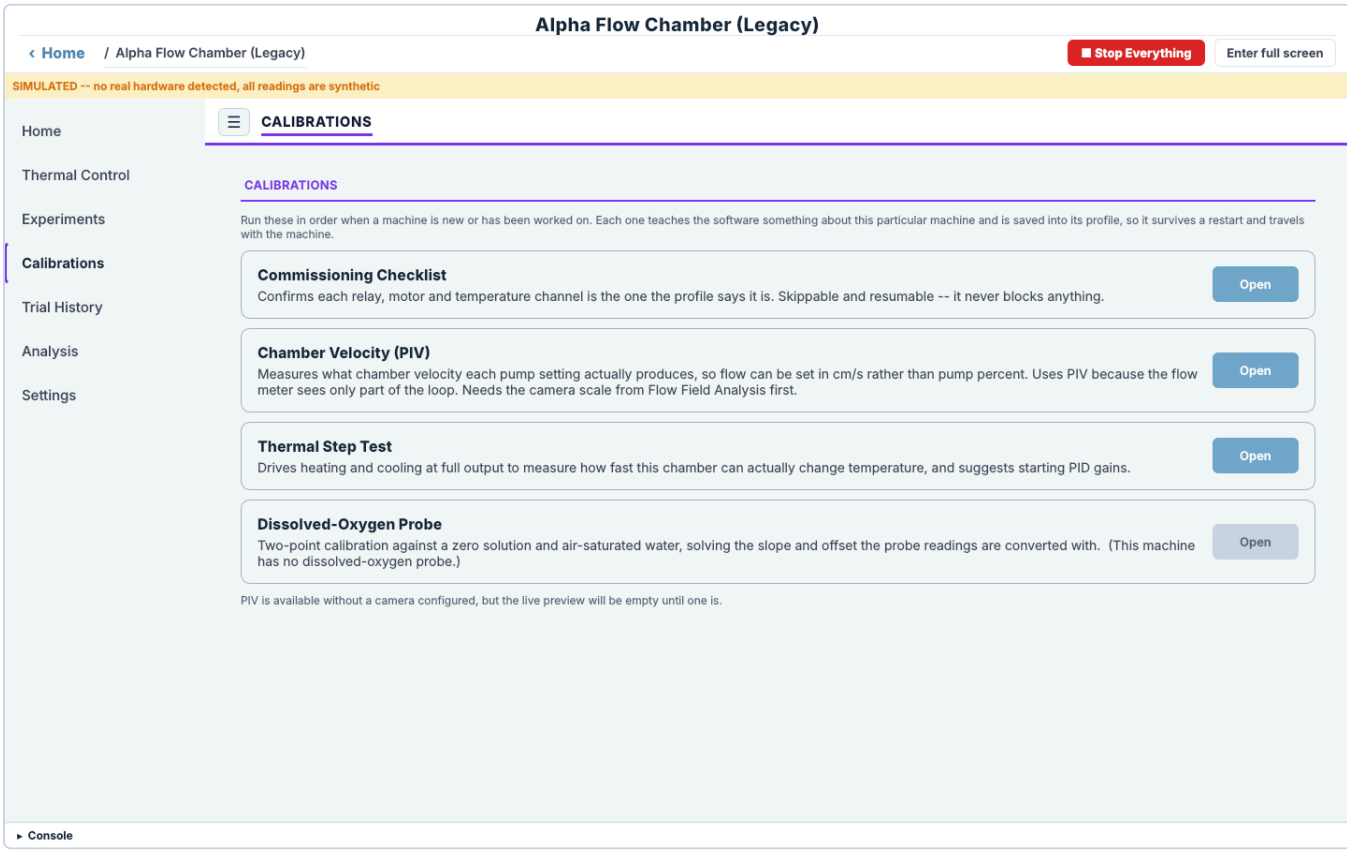
1 Commissioning a new machine

Before a machine's numbers can be trusted, the software has to be told the truth about it: which wire is which, how fast its pumps really move the water, how fast it can change temperature, and what its oxygen probe reads. This guide takes a new or rewired machine through all four, in order.

> Do this on the machine's own control computer, connected to the hardware. If the amber **SIMULATED** banner is showing, you are commissioning the simulator, not the machine.

> **Aquafab Workbench is not a safety system.** Commissioning confirms the software's description of the machine. It does not replace the machine's physical emergency stop, overheat protection or electrical checks.

Everything below is under **Calibrations** in the machine's section rail. Each result is saved into the machine's profile, so it survives restarts and updates.



The Calibrations menu.

1.1 1. Enter the machine's dimensions

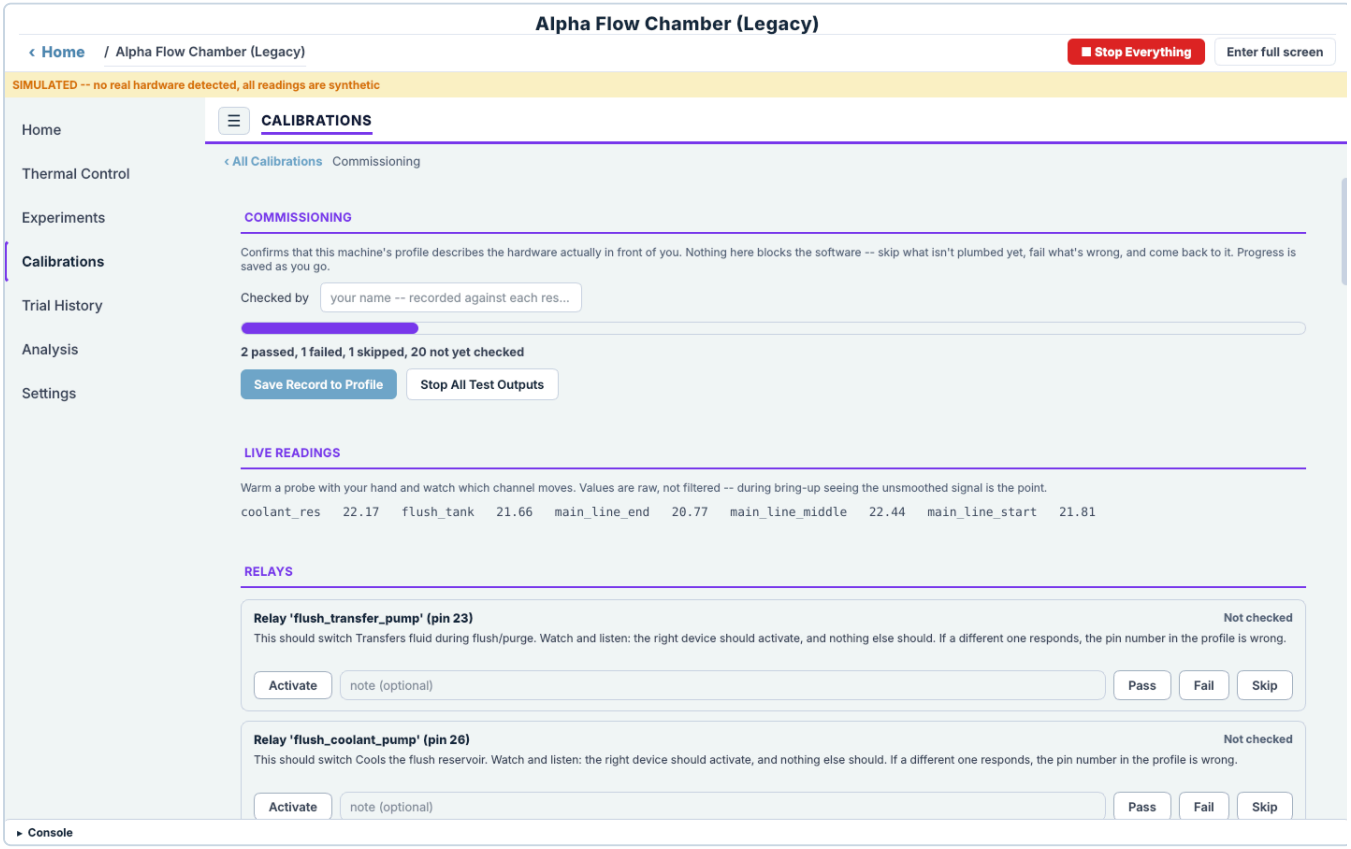
In **Settings → This machine's limits**, enter:

- **Working section volume** (L), measured. *Every respirometry rate is divided by it.*
- **Working section cross-section** (cm^2), width × height. The velocity calibration and the swim blocking correction need it.
- **Minimum** and **Maximum target temperature**, as guards against a mistyped setpoint.

Press **Save changes**.

1.2 2. The Commissioning checklist

Calibrations → Commissioning Checklist → Open.



The Commissioning checklist.

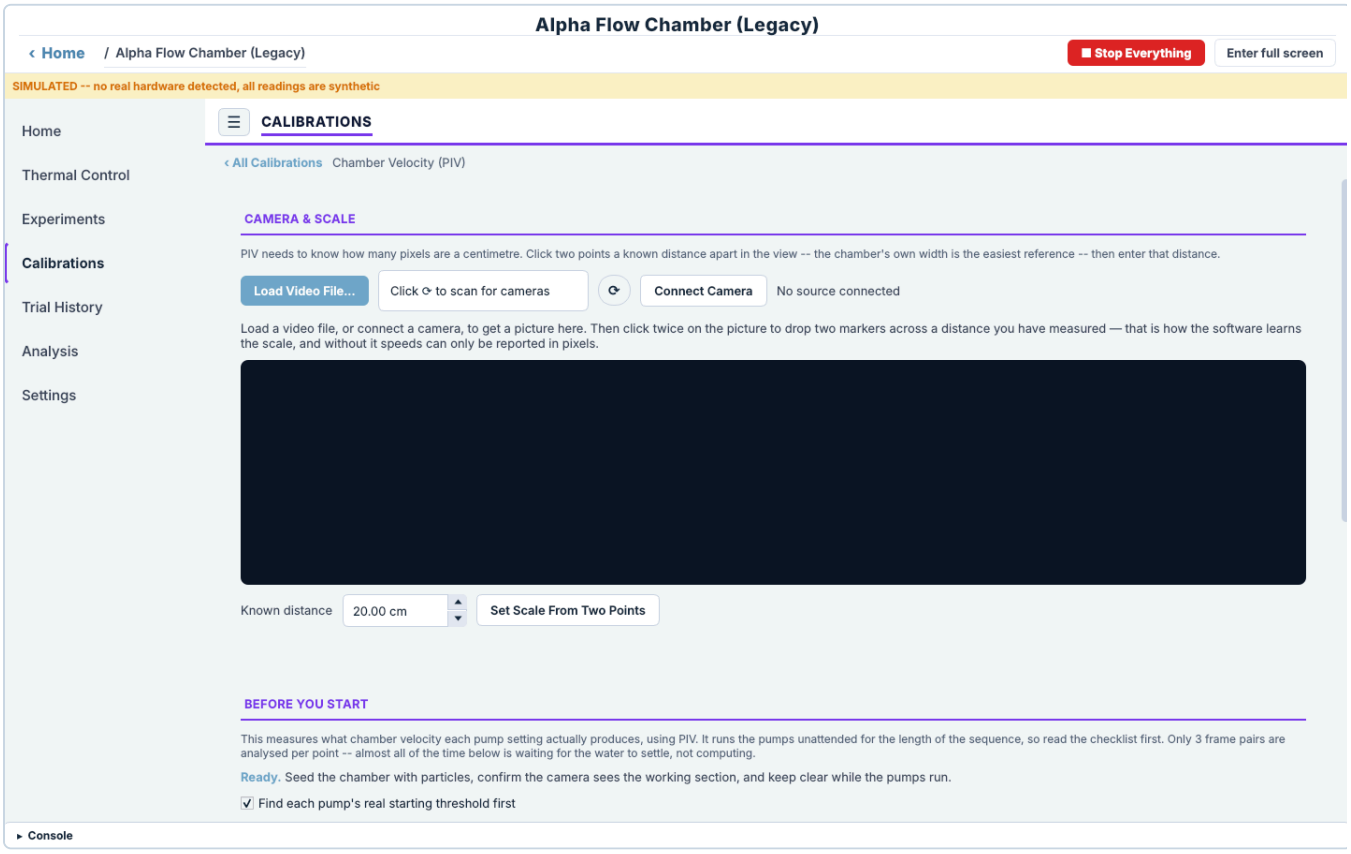
1. Type your name in **Checked by**.
2. **Relays.** For each: press **Activate**, watch and listen. The named device, and nothing else, should switch. Press **Pass**, **Fail** or **Skip**, with a note (*"clicked, heater lit"*). If a different device responds, the pin in the profile is wrong.
3. **Motors. Activate** runs the pump briefly at its declared minimum start. Confirm the right pump turns, in the right direction, and that it actually starts.
4. **Temperature probes.** Hold each probe in your hand and watch **Live readings**: only the named channel should rise. Swapped wires here make every later measurement wrong.
5. **Flow.** With the loop running, **Run Check**: the reading should be steady and plausible.
6. **Configuration.** Check the declared chiller power, reservoir volume and working section against the machine.

7. **Save Record to Profile.** Use **Stop All Test Outputs** if anything is left running.

Nothing on the checklist blocks the software: skip what is not plumbed yet and come back to it. Fix every failure in the wiring, or in the profile (**Settings → Edit Profile**), and check it again.

1.3 3. Chamber velocity, by PIV

Until this is done, a flow slider is a motor output, not a speed. It needs a camera over the working section and tracer particles in the water. **No animal in the chamber**: the pumps run unattended.



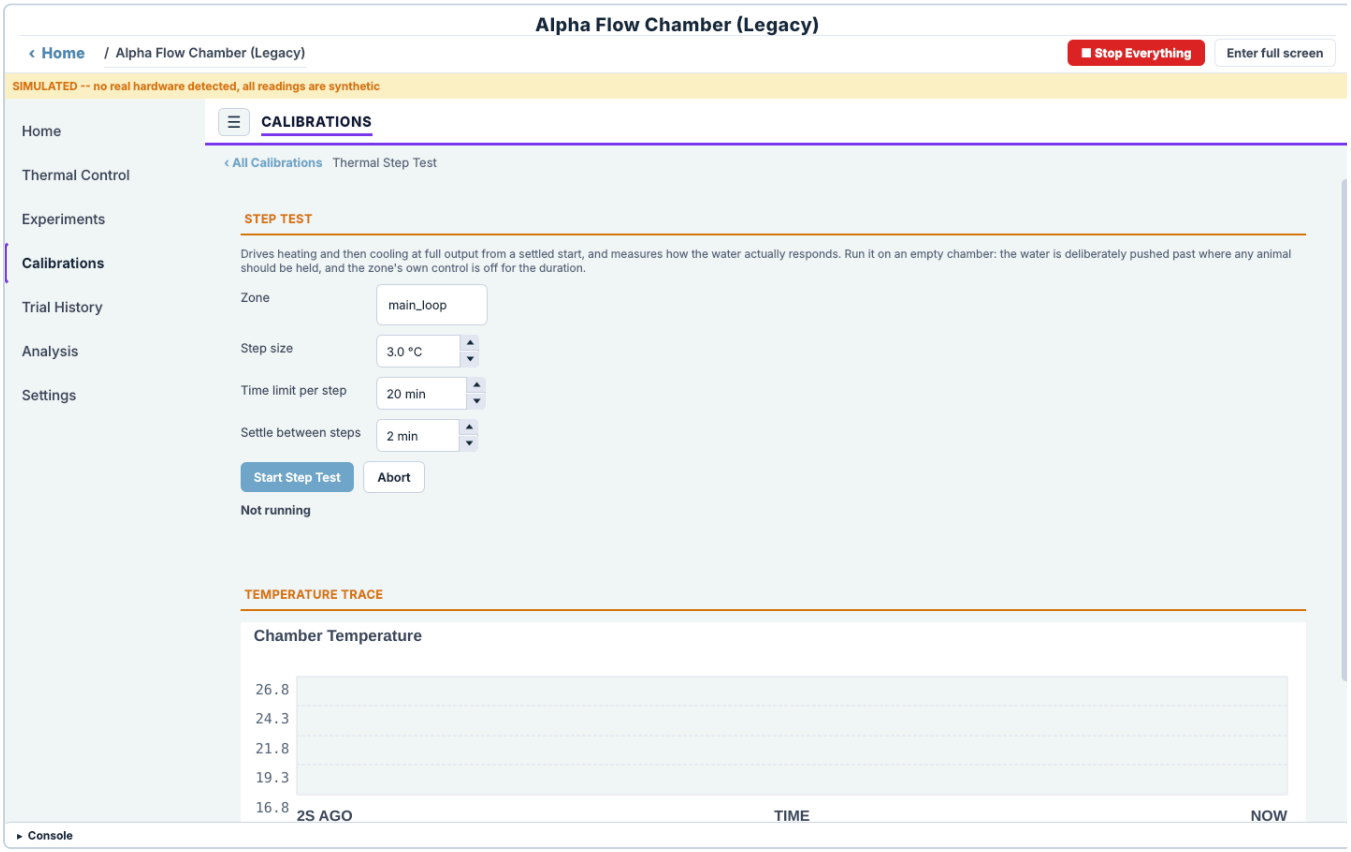
The Chamber Velocity calibration.

1. **Calibrations → Chamber Velocity (PIV) → Open.**
2. **Camera & scale**: **Connect Camera**, click two points a known distance apart (the chamber's width), type **Known distance**, press **Set Scale From Two Points**.
3. **Before you start** must say **Ready**. If it says **Not ready**, fix what it lists.
4. Leave **Find each pump's real starting threshold first** ticked.
5. Keep clear and press **Start Calibration**. Each point settles, then is measured. **Abort & Stop Pumps** stops everything.
6. Review **Measured points** (pump settings, velocity, implied flow, meter), then **Save to Profile**.

The Home page's flow slider now reads in cm/s. The *PIV Flow Measurement and Chamber Velocity Calibration* guide covers this step in more depth.

1.4 4. Thermal step test

Empty the chamber of animals first: the water is driven past where an animal should be held.



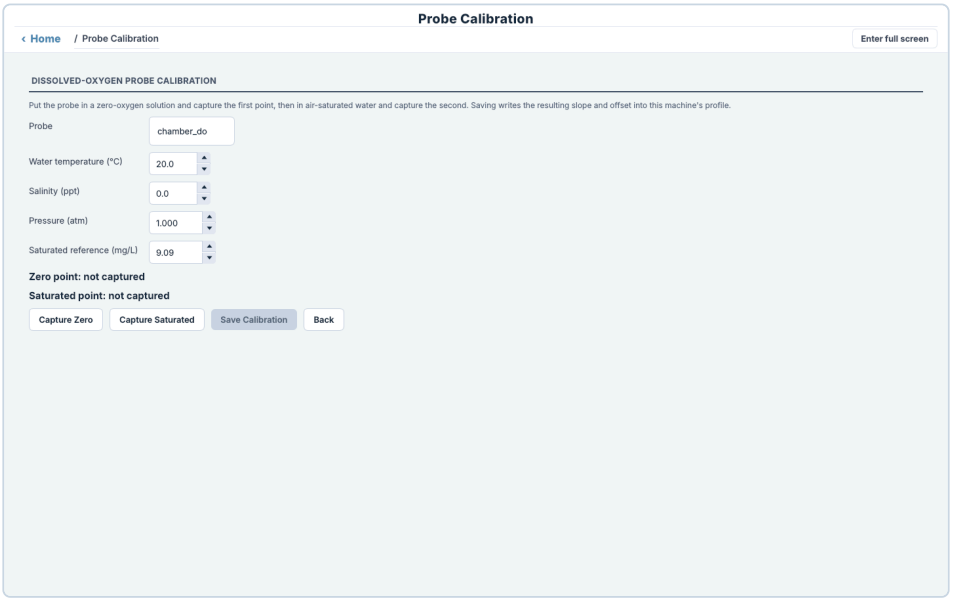
The Thermal Step Test.

1. **Calibrations → Thermal Step Test → Open.**
2. Choose the **Zone**; leave **Step size** (3 °C), **Time limit per step** and **Settle between steps** at their defaults.
3. **Start Step Test**, and leave it to run (it heats at full output, settles, then cools at full output).
4. Read the **Measured response**: heating and cooling rates (°C/min), dead time, time constant.
5. **Save to Profile.** This records the response; it does not change how the zone is controlled. The temperature-program page uses these rates to warn when a curve is steeper than the machine can follow.

> **About the suggested PID gains.** The page also suggests Kp, Ki and Kd from the Ziegler–Nichols rules. Those rules were validated only for plants whose dead time is roughly 0.1 to 1 times their time constant. Aquafab chambers are lag-dominant, far below that range, and on them the rules prescribe a Kp so high it will almost certainly oscillate; the page warns of this in amber. **Do not press Apply Gains to Zone expecting a tuned loop.** If you use PID at all, start well below the suggested Kp and raise it only if the chamber is too slow to reach its setpoint.

1.5 5. Dissolved-oxygen probe

On machines with an oxygen probe.



The oxygen probe calibration.

1. **Calibrations → Dissolved-Oxygen Probe → Open.**
2. Put the probe in a **zero-oxygen solution** (fresh sodium sulfite). When the reading is steady, **Capture Zero**.
3. Rinse, and put it in **air-saturated water**. Enter the **Water temperature**, **Salinity** (0 fresh, about 35 seawater) and **Pressure**; the **Saturated reference** is computed for you (García & Gordon 1992). When steady, **Capture Saturated**.
4. **Save Calibration.**
5. Rinse the probe thoroughly before it goes near an animal.

Recalibrate whenever the probe is new, serviced or drifting.

1.6 Done

The machine is ready. Its commissioning record, velocity calibration, thermal response and probe calibration are all stored with it, and none of them is overwritten by a software update.